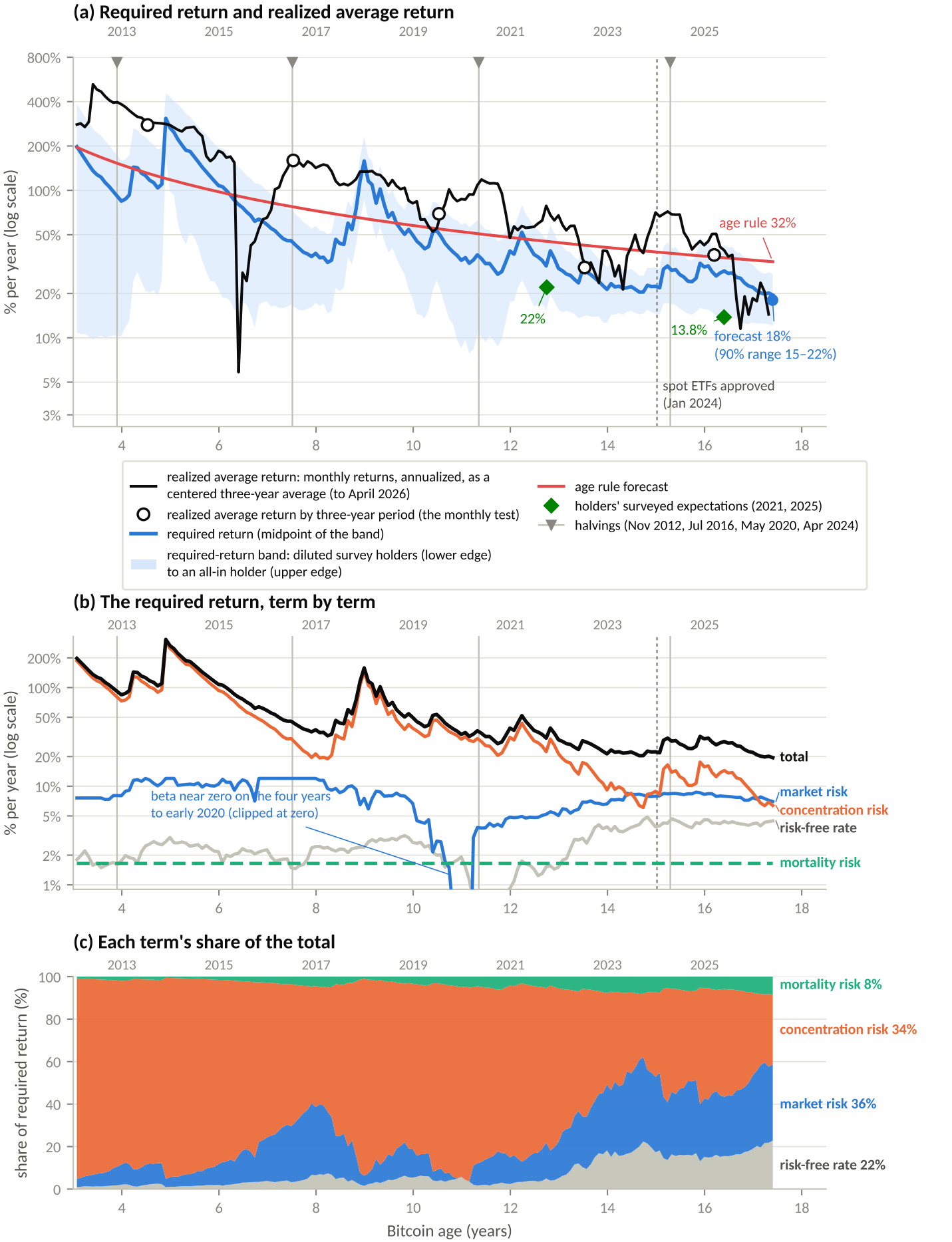

(a) Required return and realized average return
2013
2015
2017
2019
2021
2023
2025
800%
400%
200%
100%
50%
20%
10%
5%
3%
% per year (log scale)
age rule 32%
22%
13.8%
forecast 18%
(90% range 15–22%)
spot ETFs approved
(Jan 2024)
4
6
8
10
12
14
16
18
realized average return: monthly returns, annualized, as a centered three-year average (to April 2026)
realized average return by three-year period (the monthly test)
required return (midpoint of the band)
required-return band: diluted survey holders (lower edge) to an all-in holder (upper edge)
age rule forecast
holders' surveyed expectations (2021, 2025)
halvings (Nov 2012, Jul 2016, May 2020, Apr 2024)
(b) The required return, term by term
200%
100%
50%
20%
10%
5%
2%
1%
total
market risk
concentration risk
risk-free rate
mortality risk
beta near zero on the four years to early 2020 (clipped at zero)
(c) Each term's share of the total
100
80
60
40
20
0
share of required return (%)
mortality risk 8%
concentration risk 34%
market risk 36%
risk-free rate 22%
Bitcoin age (years)